

MIPS Memory Mapped I/O area

Addresses of CONTROL and DATA registers

CONTROL: .word32 0x10000

DATA: .word32 0x10008

Set CONTROL = 1, Set DATA to Unsigned Integer to be output

Set CONTROL = 2, Set DATA to Signed Integer to be output

Set CONTROL = 3, Set DATA to Floating Point to be output

Set CONTROL = 4, Set DATA to address of string to be output

Set CONTROL = 5, Set DATA+5 to x coordinate, DATA+4 to y coordinate, and DATA to RGB colour to be output

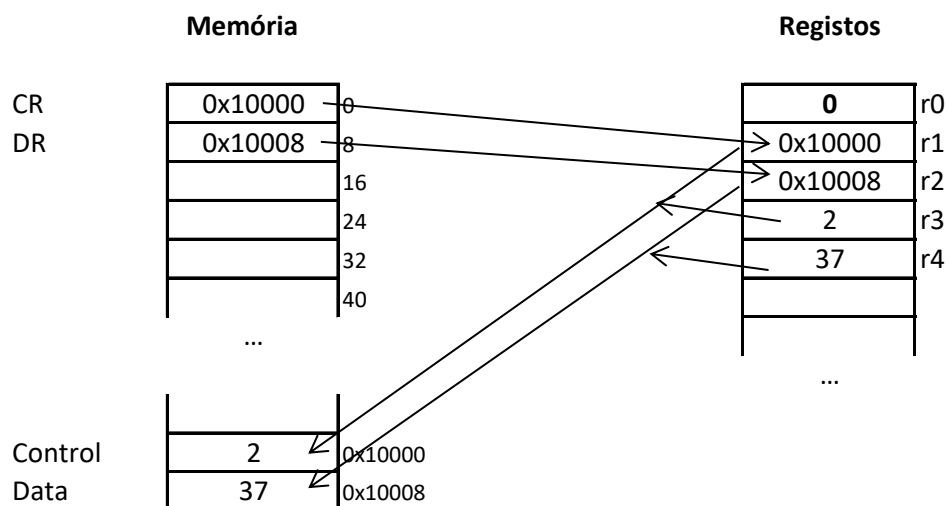
Set CONTROL = 6, Clears the terminal screen

Set CONTROL = 7, Clears the graphics screen

Set CONTROL = 8, read the DATA (either an integer or a floating-point) from the keyboard

Set CONTROL = 9, read one byte from DATA, no character echo.

Signed Integer output to screen (Mostrar r4)

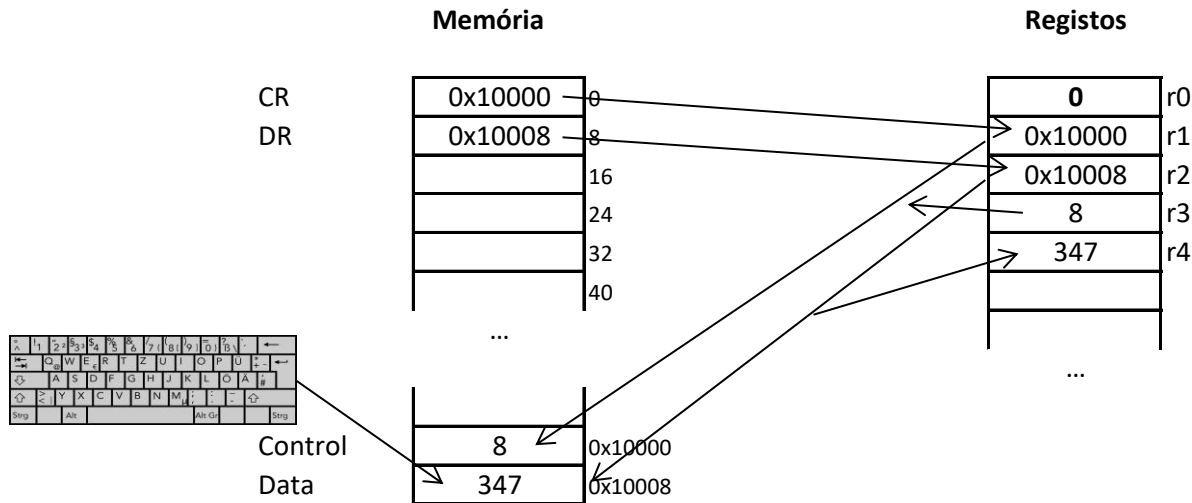


```
.data
CR:      .word32 0x10000
DR:      .word32 0x10008
```

```
.text
lwu r1,CR(r0) ;Control Register
lwu r2,DR(r0) ;Data Register
```

```
sd r4,(r2)
daddi r3,r0,2
sd r3,(r1)
```

Read an integer (or floating-point) from the keyboard (Ler r4)

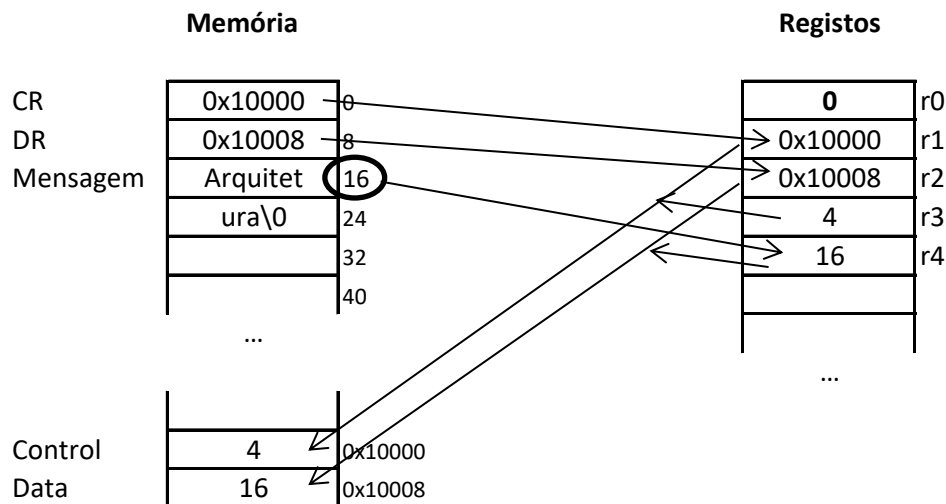


```
.data
CR:      .word32 0x10000
DR:      .word32 0x10008
```

```
.text
lwu r1,CR(r0) ;Control Register
lwu r2,DR(r0) ;Data Register
```

```
daddi r3,r0,8
sd r3,(r1)
ld r4,(r2)
```

String output to screen (Mostrar Mensagem)

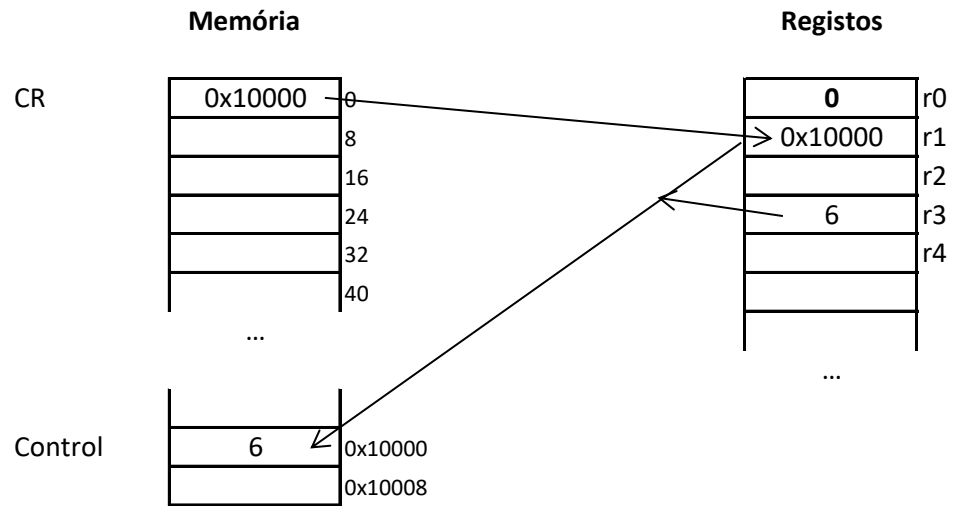


```
.data
CR:      .word32 0x10000
DR:      .word32 0x10008
Mensagem: .asciiz "Arquitetura"
```

```
.text
lwu r1,CR(r0) ;Control Register
lwu r2,DR(r0) ;Data Register
```

```
daddi r4,r0,Mensagem
sd r4,(r2)
daddi r3,r0,4
sd r3,(r1)
```

Clears the terminal screen (Apaga ecrã)



```
.data
CR:      .word32 0x10000

.text
lwu r1,CR(r0) ;Control Register

daddi r3,r0,6
sd r3,(r1)
```